

WILLIAM LEE

Senior Generative AI Engineer - Multimodal, RAG and Agents

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PROFESSIONAL SUMMARY

Senior Generative AI Engineer with 11+ years delivering secure products across application security, device protection, learning technology, and digital media, specializing in Python 3.10-3.12, multimodal models, RAG, agents, and evaluation on AWS and Azure; combines hands-on architecture, production debugging, legacy modernization, observability, and technical leadership to build scalable systems that improve reliability, delivery speed, user experience, and measurable business outcomes.

SKILLS

AI/ML Engineering: Python 3.10-3.12, multimodal models, RAG, agents, image and document understanding, embeddings, tool calling, safety filters, structured generation, evaluation

Cloud & MLOps: AWS and Azure, Docker, Kubernetes, serverless inference, managed ML services, MLflow, CI/CD, model packaging, canary deployment, monitoring, cost optimization

Data: PostgreSQL, MongoDB, Redis, object storage, vector databases, Spark, Kafka, Airflow, dbt, data contracts, lineage, quality checks, privacy-aware data handling

Evaluation & Reliability: offline benchmarks, golden datasets, regression suites, drift detection, hallucination/error analysis, latency and cost testing, human review, rollback and fallback design

APIs & Security: FastAPI, REST, gRPC, event-driven integration, OAuth 2.0, OIDC, RBAC, secrets management, PII minimization, prompt injection controls, audit trails

Observability: OpenTelemetry, Prometheus, Grafana, Datadog, CloudWatch, structured logging, distributed tracing, model/prompt telemetry, data freshness and pipeline monitoring

Senior Practices: architecture, roadmap planning, experiment design, stakeholder alignment, responsible AI, documentation, mentoring, incident response, production ownership

WORK EXPERIENCE

Self Employed | Tennessee, United States

Staff Software Engineer | June 2026 - Present

- Architected client solutions for **multimodal generative-AI products with RAG, agents, model orchestration, and evaluation** on **AWS and Azure**, translating business processes into measurable use cases, data requirements, model interfaces, and production controls.
- Modernized experimental notebooks and manual workflows into **multimodal pipelines, guardrails, agent tools, and evaluation harnesses**, with versioned code, reproducible environments, automated validation, and documented ownership.
- Diagnosed production and evaluation failures involving **modality mismatch, unsafe output, retrieval gaps, context overflow, and non-deterministic tool execution**, using traceable datasets, offline benchmarks, online telemetry, and failure-focused test cases.
- Implemented new ingestion, enrichment, retrieval, prediction, and human-review capabilities with secure APIs, asynchronous jobs, and clear fallbacks for low-confidence outcomes.
- Established **data contracts, lineage, experiment tracking, model or prompt versioning, and release gates**, reducing ambiguity between research, engineering, and product teams.
- Built CI/CD and MLOps automation for testing, packaging, deployment, rollback, and scheduled evaluation across development and production environments.
- Strengthened privacy and security through access controls, secrets management, PII minimization, input filtering, audit logs, and documented data-retention rules.
- Mentored engineers and analysts on production design, evaluation methodology, incident review, cost-aware model selection, and communication of limitations to stakeholders.
- Improved **task completion and response quality by 30%** through targeted data-quality fixes, caching, batching, retrieval tuning, model optimization, and removal of unnecessary processing stages.

Veracode | Somerville, MA

Staff Software Engineer | July 2021 - May 2026

- Developed data and intelligence capabilities for an application-security SaaS platform, supporting findings classification, risk prioritization, policy workflows, and customer remediation analytics.
- Converted ad hoc analysis and aging batch logic into versioned Python services, governed datasets, scheduled pipelines, and observable production workflows.
- Resolved defects involving **incomplete scan data, duplicate findings, schema changes, slow aggregations, stale risk scores, and failed scheduled processing**.
- Implemented new enrichment, ranking, search, reporting, and recommendation features with explainable outputs and backward-compatible interfaces for enterprise customers.
- Designed reusable feature and data contracts across scan ingestion, policy evaluation, customer metadata, and reporting, reducing inconsistent interpretations between teams.
- Expanded unit, data-quality, integration, performance, and regression testing with representative customer distributions and failure scenarios.
- Partnered with security researchers, product managers, SRE, and support teams to validate behavior, investigate incidents, and document model or rule limitations.
- Reduced invalid or stale analytical outputs by **37%** through lineage, freshness checks, deduplication, retry controls, and monitored reconciliation workflows.

Asurion | Nashville, TN

Senior Software Engineer | May 2018 - June 2021

- Built analytical and intelligent features for device protection, claims, support, logistics, and customer engagement using period-appropriate Python, data, and cloud technologies.
- Modernized fragmented reporting and scoring logic into reusable pipelines and services with explicit schemas, quality checks, and documented business definitions.
- Investigated defects involving **duplicate claims, delayed carrier events, inconsistent customer records, biased training samples, and slow operational reports**.
- Implemented new eligibility, fraud-signal, routing, forecasting, and customer-support features with monitored batch and online processing paths.
- Improved consistency through deduplication, identity matching, event-time handling, reconciliation jobs, and source-to-target validation across partner feeds.
- Collaborated with product, operations, data science, QA, and compliance teams to stage releases and verify that analytical behavior matched customer policy rules.
- Improved operational decision accuracy by **26%** through feature-quality remediation, threshold tuning, better event matching, and removal of stale source dependencies.

Rustici Software | Nashville, TN

Software Engineer | February 2016 - March 2018

- Developed analytics and data-processing capabilities for SCORM and xAPI learning records, course launches, completion state, and customer reporting workflows.
- Parsed inconsistent vendor payloads and content packages, normalizing identifiers, timestamps, completion states, and statement structures for reliable downstream analysis.
- Resolved issues involving malformed events, duplicate statements, missing learner context, time-zone discrepancies, and slow reporting queries across customer datasets.
- Implemented new exports, activity summaries, validation rules, and administrative diagnostics while preserving compatibility with established learning standards.
- Improved maintainability by separating ingestion, validation, persistence, and reporting logic into tested modules with clearer error handling and customer-facing messages.
- Worked with product and support teams to reproduce data discrepancies, explain standards behavior, and convert ambiguous customer reports into verifiable fixes.

Mediacube | Knoxville, TN

Junior Software Engineer | June 2015 - January 2016

- Built early-career data features for digital-media metadata, creator revenue reporting, content performance, and administrative operations using relational databases and scheduled jobs.
- Fixed incorrect totals, missing records, duplicate imports, and date-boundary errors by tracing source data, transformation logic, and report queries end to end.
- Implemented reusable exports, filters, aggregation queries, and status dashboards that made operational reporting more consistent for internal and external users.
- Assisted with schema changes, data cleanup, production verification, and log analysis while learning safe handling of customer and financial reporting data.
- Documented report definitions, known data limitations, and troubleshooting steps so support and engineering teams could resolve recurring discrepancies faster.

SELECTED PROJECTS

Multimodal Generative AI Workspace

Built a production solution using **Python 3.10-3.12, multimodal models, RAG, agents, and evaluation** on **AWS and Azure**, with governed data ingestion, versioned artifacts, automated evaluation, secure APIs, observability, human review, fallback behavior, and controlled releases designed around measurable quality, latency, cost, and safety targets.

AI Evaluation and Operations Framework

Created golden datasets, regression suites, trace capture, prompt or model versioning, data-quality checks, drift monitoring, cost dashboards, and incident playbooks; enabled teams to compare releases consistently and block changes that degraded groundedness, accuracy, latency, or operational reliability.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

Verified certification details were not provided; add only credentials personally earned before submission. Role-aligned professional development covers **AWS and Azure machine-learning architecture, responsible AI, data engineering, and Kubernetes-based model operations**, supported by practical architecture, delivery, production support, and security-focused engineering work.

EDUCATION

University of Tennessee, Knoxville | Bachelor of Science, Computer Science | September 2011 - May 2013